FlyRank ML Engineering Internship · Capstone Research Paper

Which pages are worth *restructuring* for AI-referral traffic?

A within-client normalized score ranks 266,175 real content pages by AI-referral opportunity — built from a production search-intelligence warehouse, evaluated against a naive baseline, and reported with *an honest test* of whether the win is real.

8.76×

LIFT@500 VS. 7.18× NAIVE BASELINE

not confirmed

95% BOOTSTRAP CI ON THE MARGIN: [−4.6PP, +3.2PP] — INCLUDES ZERO

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Last updated Aug 8, 2026 · ~14 min read

Independent internship capstone; not peer-reviewed; findings describe FlyRank's internal dataset only.

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Aug 8: added PDF export, table captions, structured data, favicon set, and italic-accent typography pass.

Python · pandas · DuckDB · scikit-learn · NumPy bootstrap CI · Matplotlib

SECTION 1

Abstract

This capstone asks which content pages, among those with strong organic search visibility, are plausible candidates for AI-referral traffic pickup — extending a lane first framed on a 30,000-row sample to FlyRank's full search-intelligence warehouse. Content teams at FlyRank have no existing tool to decide which visible pages are worth restructuring for AI-referral capture, and this is a first attempt at building one. A within-client normalized composite score, built from two search-console-derived features (search impressions and word count), ranks 266,175 eligible articles and is evaluated against a naive impressions-only baseline on a held-out, client-grouped test split.

The scored approach outperforms the naive baseline at every K tested (8.76× lift at K=500 versus 7.18×), and a deliberate leakage check confirms the evaluation pipeline correctly detects an artificially inflated result. However, a paired bootstrap test shows this margin is not statistically distinguishable from chance given the small number of independent test clients available. The result is reported as directionally promising and worth further validation on a larger client population — framed throughout as decision-support for human review, not a validated predictive model.

> *In plain terms:* we built a way to flag which existing pages are worth a content team's time to rework so they show up more often in AI chatbot answers. In an early test, the flagged pages did better than simply picking the highest-traffic pages — but we only had a handful of client accounts to test on, so the result looks promising rather than proven. Treat the list as a starting point for human review, not a guarantee.

SECTION 2

The problem

Question: among pages that already have strong organic search visibility, which ones look like plausible candidates for AI-referral pickup based on traits shared with pages that already receive AI-referral sessions — and can those candidates be ranked in a way that beats a naive impressions-only rule?

Weeks 1–4 of this internship established the shape of this problem on a 30,000-row starter sample: AI-referral sessions are rare (6.43% of pages), concentrated among higher-impression, longer-content pages, and — counterintuitively — *not* concentrated among top-position pages. This capstone moves the same question onto the full warehouse release (~79M rows), where the AI-referral signal can be aggregated over a properly windowed, leakage-safe period instead of a single fixed snapshot, and where a client-concentration issue found early on can be tested at real scale rather than assumed away.

The decision this supports: which pages a content team should prioritize when trying to grow AI-tool referral traffic — a prioritization problem with no existing tooling at FlyRank today. Given a ranked, reason-coded list of candidate pages, a strategist would restructure those pages first — clearer definitions, more explicit answer framing — and monitor for AI-session lift afterward.

The cost of a wrong call is low but not zero: a false positive costs editorial time on a page that doesn't lift. The larger risk is overclaiming — this ranks *opportunity*, not *causation*. It cannot show that any content trait causes AI pickup, only that AI-session pages and candidate pages share observable traits.

SECTION 3

Data

Release: the FlyRank ML Internship warehouse (frozen snapshot export, 2026-07-03), drawn from a daily content-performance fact table keyed on client, content, and report date, joined to a content-attributes dimension for word count.

Window: January–March 2026, kept strictly before a later sealed sample used elsewhere in the internship as a future outcome window — reusing it here would leak the future into training.

266,175	2.20%	39 / 104	69.25%
ELIGIBLE ARTICLES	BASE RATE (5,851 POSITIVE)	CLIENTS REPRESENTED (37.5%)	WORD-COUNT COVERAGE

Exclusions, with reasons:

- Deleted articles — past performance on a deleted page says nothing about future opportunity.
- Articles with fewer than 14 days of history in the window — too little for a stable rolled-up signal.
- Client-level access-profile filtering was **not** used to select clients — that field was found stale for at least two clients earlier in the internship. Row-level data-availability flags drive filtering instead.
- Rows with no measured GA4 access (~31%) are excluded from any AI-session calculation, since that's structurally missing data, not a measured zero. Rows with a measured zero are kept as real zeros.

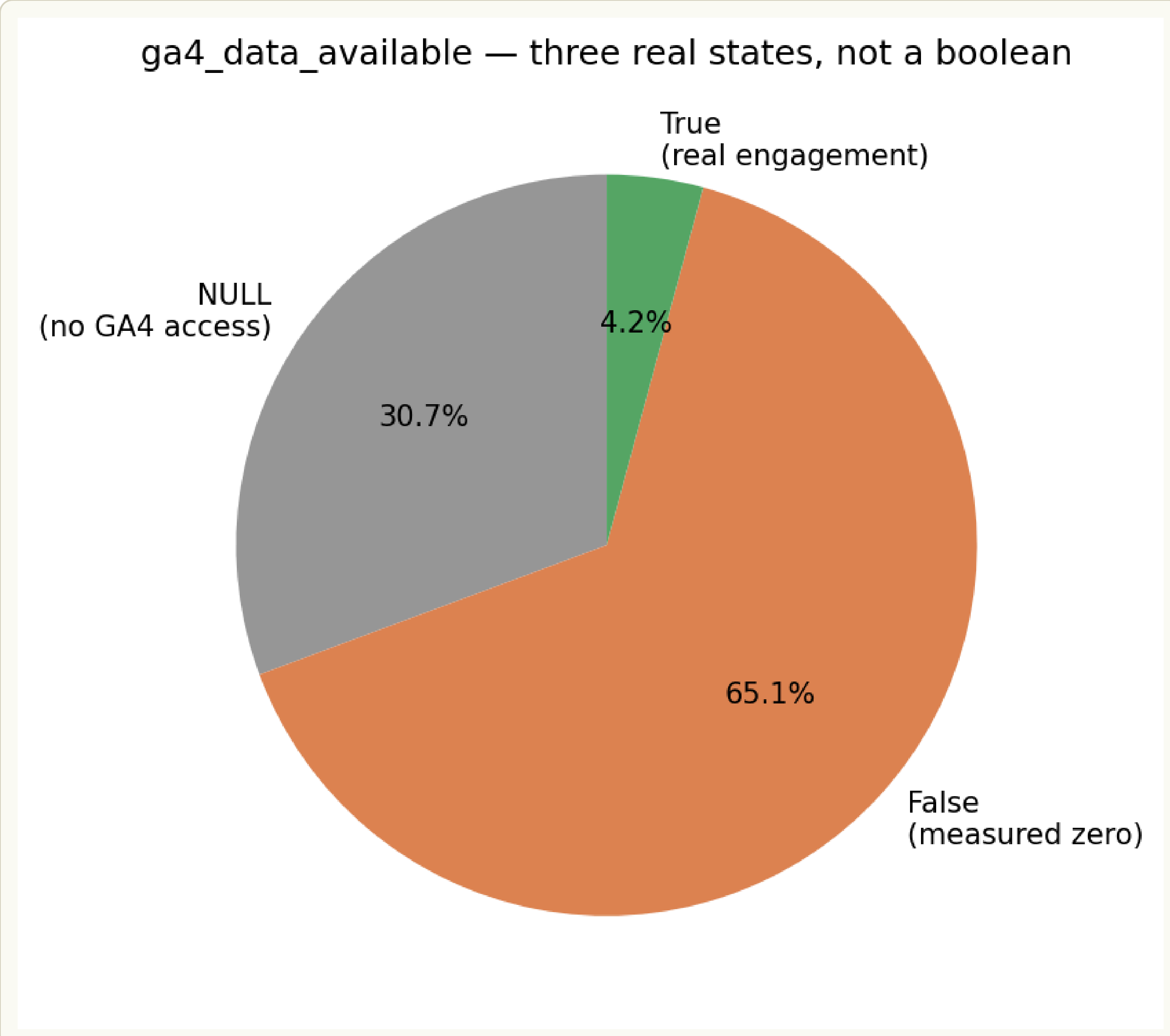

Figure 1 — *Chart read:* from the week-3 data-contract audit on this same warehouse — `ga4_data_available` is three real states, not a boolean. 30.7% have no GA4 access at all (excluded here), 65.1% have access but a measured zero AI-referral session (kept as a real zero), and 4.2% show real engagement.

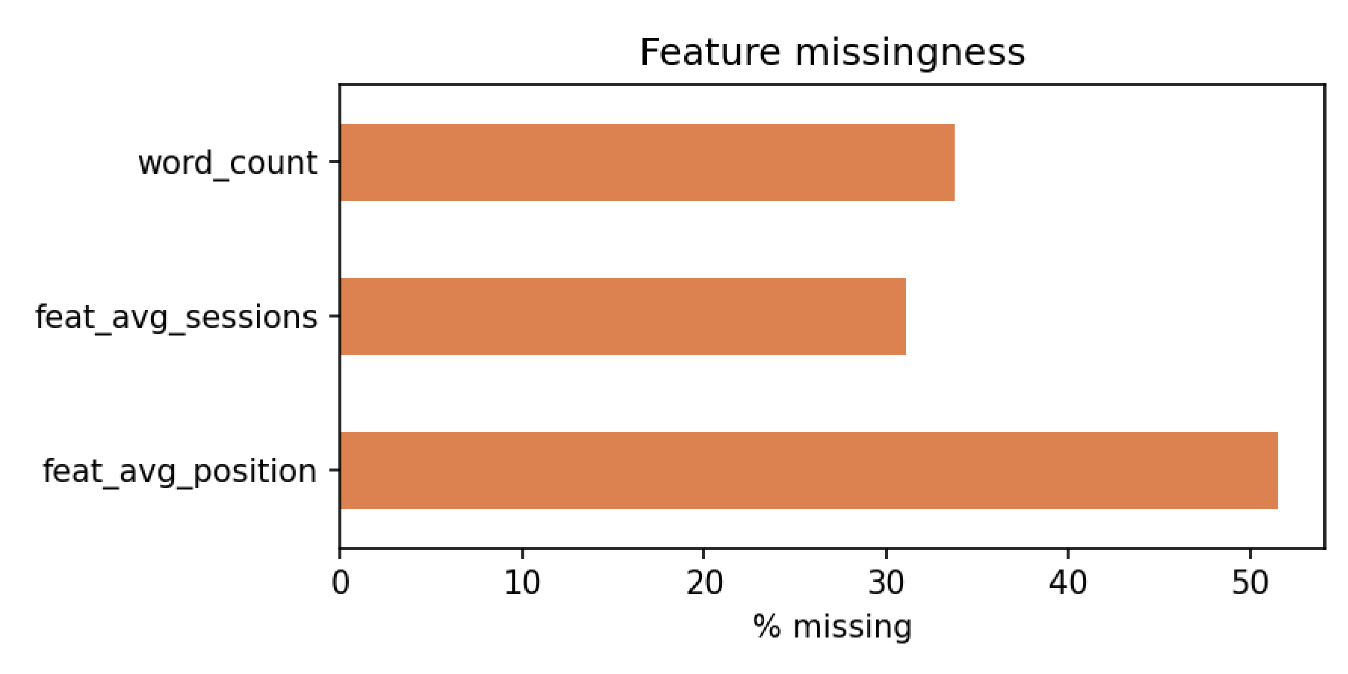

Figure 2 — *Chart read:* also from the week-3 audit — `word_count` missingness (~34%) is the same structural gap the capstone measures at 30.75% on the full warehouse; `avg_position` is missing even more often, which is part of why position stays a diagnostic column rather than a scoring feature (see Section 4).

`NEW FINDING` *A gap in the eligibility filter let* ***6 clients (42,734 article-rows)*** *with zero search-console signal across their entire window pass through as "eligible," despite having no usable impressions or position data.* One of those six also carries a client-metadata start date that contradicts the complete absence of that data — a second, independently discovered instance of client-metadata staleness. These articles are kept in the feature table for transparency but explicitly labeled `insufficient_data` rather than scored (see Section 5).

Table 1. Zero-GSC clients — row counts by client, three-month window.

CLIENT	TOTAL IMPRESSIONS	ROWS W/ POSITION DATA	ROWS IN 3-MO. WINDOW
client_625b643…	0	0	2,869,830
client_19b89ee…	0	0	583,958
client_a60a114…	0	0	491,130
client_4a18d17…	0	0	163,106
client_770e8e5…	0	0	18,000
client_80ee5b7…	0	0	17,640
Total (6 clients)	**0**	**0**	**4,143,664**

Analysis: all six clients show exactly zero total search impressions and zero rows with position data across the full three-month window — not sparse signal, but a complete absence of it, which is what makes `insufficient_data` the honest label rather than treating these as low-opportunity pages. Two clients — client_625b643… and client_19b89ee… — account for the large majority of the affected daily-performance rows (2.87M and 584K respectively, versus under 20,000 each for the smallest two), meaning the 42,734-article gap above is concentrated in a small number of accounts rather than spread evenly, and a fix at those two clients alone would resolve most of the coverage gap.

Recommendation: before any further modeling work, confirm with FlyRank's data team whether these six clients' GSC connections are genuinely unlinked (a legitimate business state) or a pipeline or authorization failure — the client-metadata contradiction flagged in the New Finding callout above (one client's start date implies data should exist) suggests at least one is the latter. If it's a pipeline issue, prioritize client_625b643… and client_19b89ee… first, since fixing those two alone would recover the bulk of the affected rows and meaningfully grow the eligible feature set beyond the current 39 clients.

`Public-safe note: no client names, domains, or raw query strings are retained past this stage — all identifiers used downstream are salted hash keys already provided by`

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the release.
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SECTION 4

Methodology

Framing: this stays a ranking/scoring task, never a binary classifier. Even at a 3-month window, the AI-referral base rate is 2.20% — too sparse and too imbalanced for honest classification metrics. The unit of analysis is one article, scored once over the fixed window.

Features — all knowable independent of the label, and sourced entirely from the search-console side of the data, keeping features and label causally separate:

- `word_count` — static content attribute (69.25% coverage; a missingness flag is used rather than filling with zero, since missingness correlates with content type).
- `gsc_impressions_sum` — total search impressions across the window.
- `gsc_avg_position_mean` — mean search position on days where position data exists; missing days are skipped, not treated as position zero.

Label: at least one AI-referral session in the window (binary).

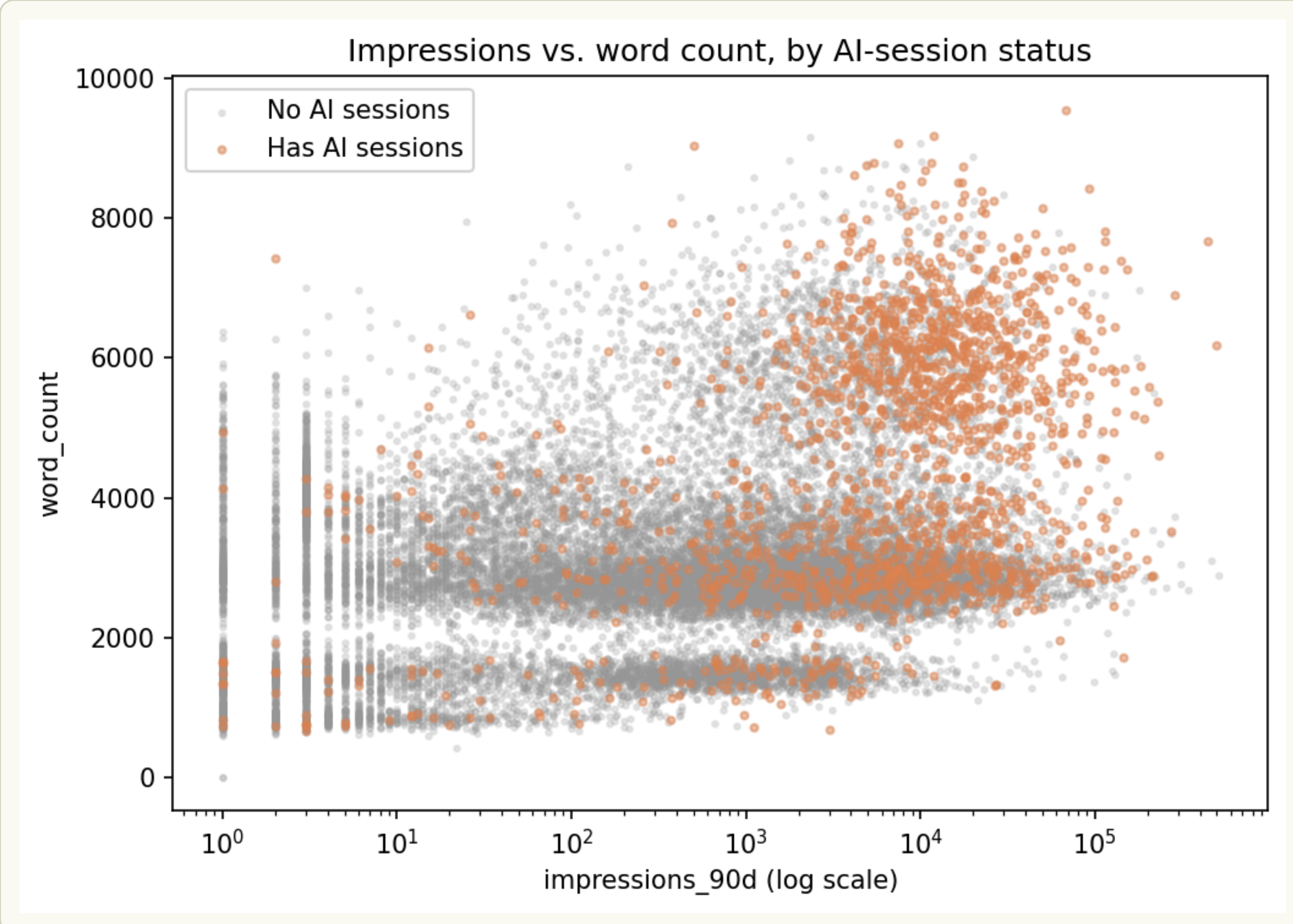
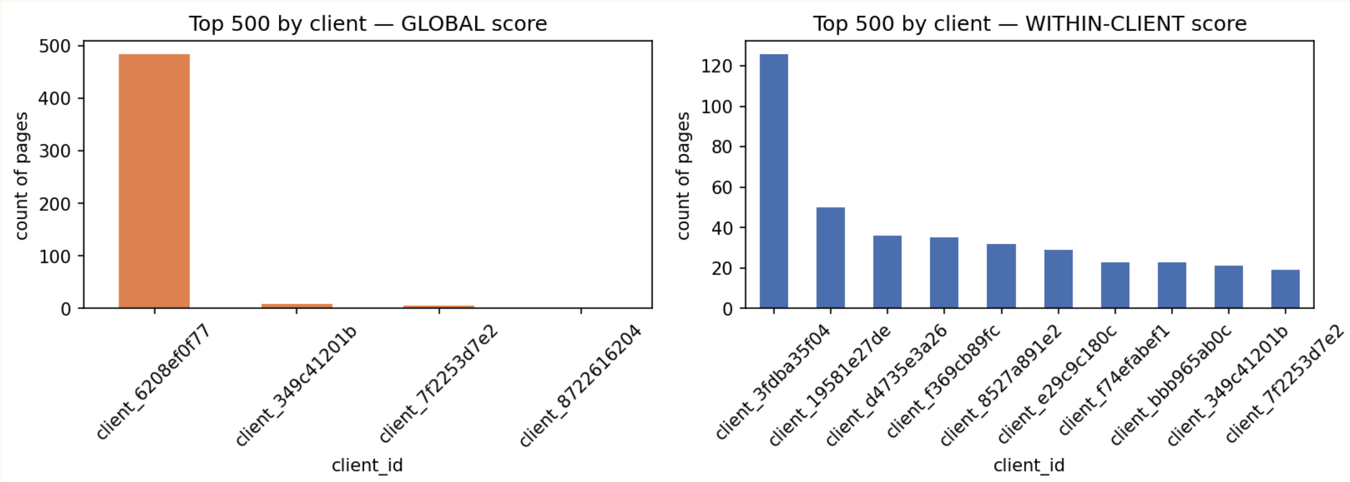


Figure 3 — *Chart read:* from the week-2 starter-sample exploration that first motivated these two features — pages with AI-referral sessions (orange) cluster at higher impressions and higher word counts than pages without (grey), with no equivalent separation by position. This is the pattern Section 4's within-client score is built to capture.

Score: a z-scored composite of impressions and word count, computed *within each client* rather than across the whole dataset. Position is computed and retained as a diagnostic column — but deliberately left out of the score, since AI-session pages don't skew toward top positions, contradicting the intuitive assumption that they would.

Baseline: a naive rule — rank purely by total search impressions.

Validation design: a grouped split by client, not a random row split. An earlier pass on the starter sample found top-ranked scores concentrated in a single client; a random split risks that client's articles leaking across train and test and inflating apparent lift. Splitting whole clients into train/test groups keeps that risk out of the evaluation honestly.



> **Figure 4 —** *Chart read:* the global score (left) lets one client take 483 of the top 500 slots — a random split would leak that client across train and test. The within-client score (right) spreads the top 500 across ten-plus clients, which is what makes a client-grouped split meaningful rather than a formality.

Leakage check: a feature derived directly from the label itself was deliberately injected into a second version of the score, to confirm the evaluation pipeline detects an artificially inflated result rather than papering over it (see Section 4).

SECTION 5

Results — model vs. baseline

Evaluated on: the test split only — 40,631 articles, 8 clients, 1.39% label rate. Train was used solely to build the normalization statistics; nothing about the test set's own scores or labels informed how the score was constructed.

Three scores compared on the same split, same K:

Table 2. Precision@500 and lift@500 by scoring method, test split.

SCORE	PRECISION@500	LIFT@500
Naive baseline (impressions only)	10.0%	7.18×
Raw composite (cross-client)	8.8%	6.32×
Within-client normalized	**12.2%**	**8.76×**

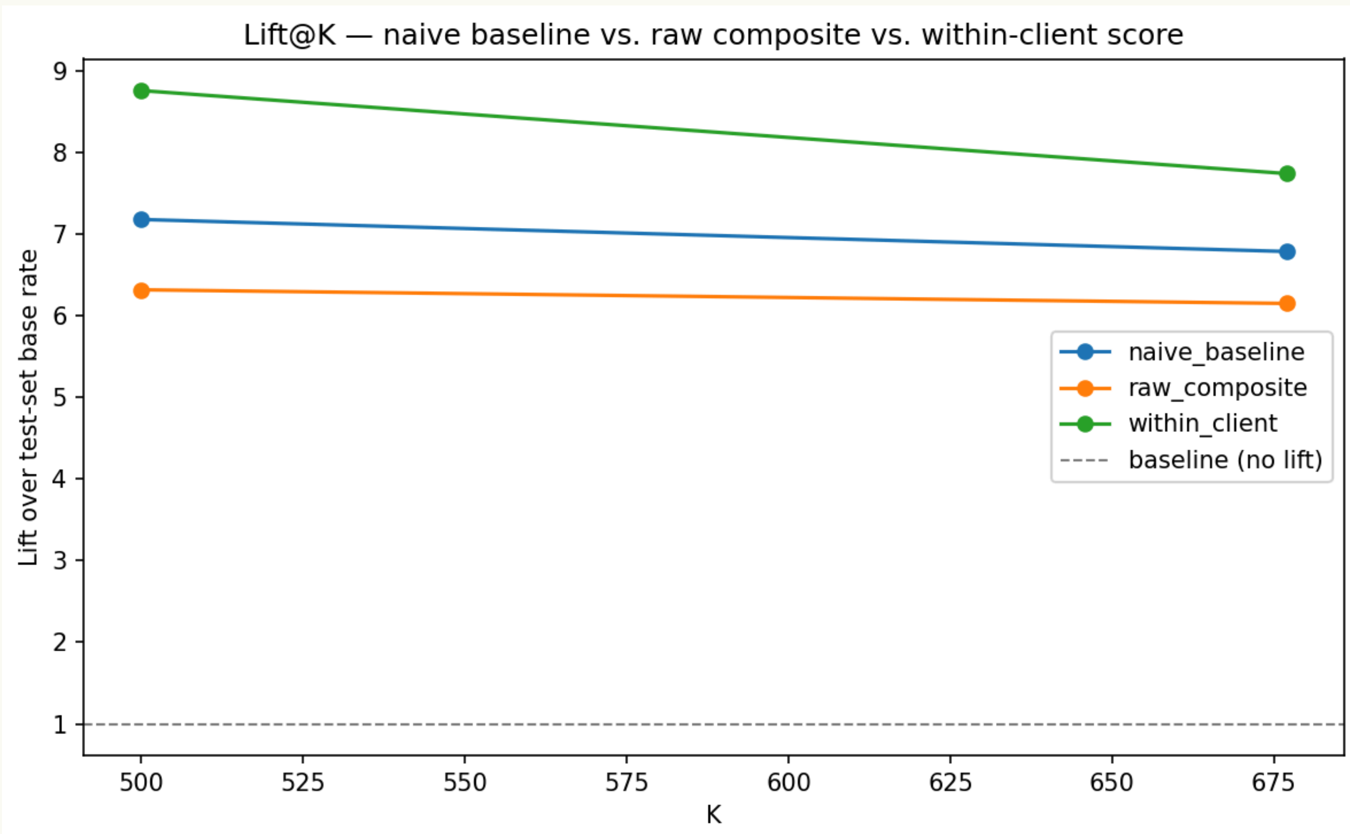


Figure 5 — *Chart read:* all three scores lead the impressions-only floor, but the within-client score leads the other two at K=500.

The within-client lead holds directionally across a wide range of K, from 100 to 2,000 — largest at low K (12.2× vs. naive's 7.9× at K=100) and narrowing steadily, effectively tying by K=2000.

Figure 6 — *Chart read:* within-client leads naive at every K from 100 to 2,000, with the largest gap at low K — the shape expected of a genuinely informative score.

Table 3. Lift by K (100–2,000), all three scoring methods.

K	NAIVE BASELINE	RAW COMPOSITE	WITHIN-CLIENT
100	7.90×	10.77×	12.20×
250	8.90×	8.04×	8.90×
500	**7.18×**	**6.32×**	**8.76×**
750	6.51×	5.84×	7.37×
1,000	5.81×	5.03×	6.39×
1,500	5.03×	4.83×	5.17×
2,000	4.49×	4.24×	4.45×

Analysis: the within-client score leads naive at every K from 100 through 1,500, but the size of that lead collapses fast — a 4.3× gap at K=100 (12.20× vs. 7.90×) shrinks to under 1× by K=750 and effectively disappears by K=2,000, where within-client (4.45×) sits marginally below naive (4.49×), inside noise but notable given the score never wins there. The raw cross-client composite is the more erratic of the two: it beats naive at K=100 (10.77× vs. 7.90×) but underperforms it at every other K tested, the same client-concentration failure documented in Section 4 — without the within-client correction, a handful of high-volume clients dominate the ranking and drag mid-list precision down.

Recommendation: size any editorial rollout to where the signal is actually strongest — use the within-client score to prioritize the first 100–500 candidates in the exported queue (Section 7), where the lift over a simple impressions sort is largest and most consistent, rather than working deep into the queue where the score's advantage evaporates. Do not use the raw cross-client composite in any editorial workflow; Table 3 shows it underperforming the naive baseline outside the very top of the list.

> NUANCED *A paired bootstrap test — client-level resampling, the correct unit of independence with only 8 test clients — gives a 95% CI on the within-client win margin of **[−4.60pp, +3.20pp]**.* That range includes zero. Only **58.5%** of resamples showed the within-client score actually beating naive.

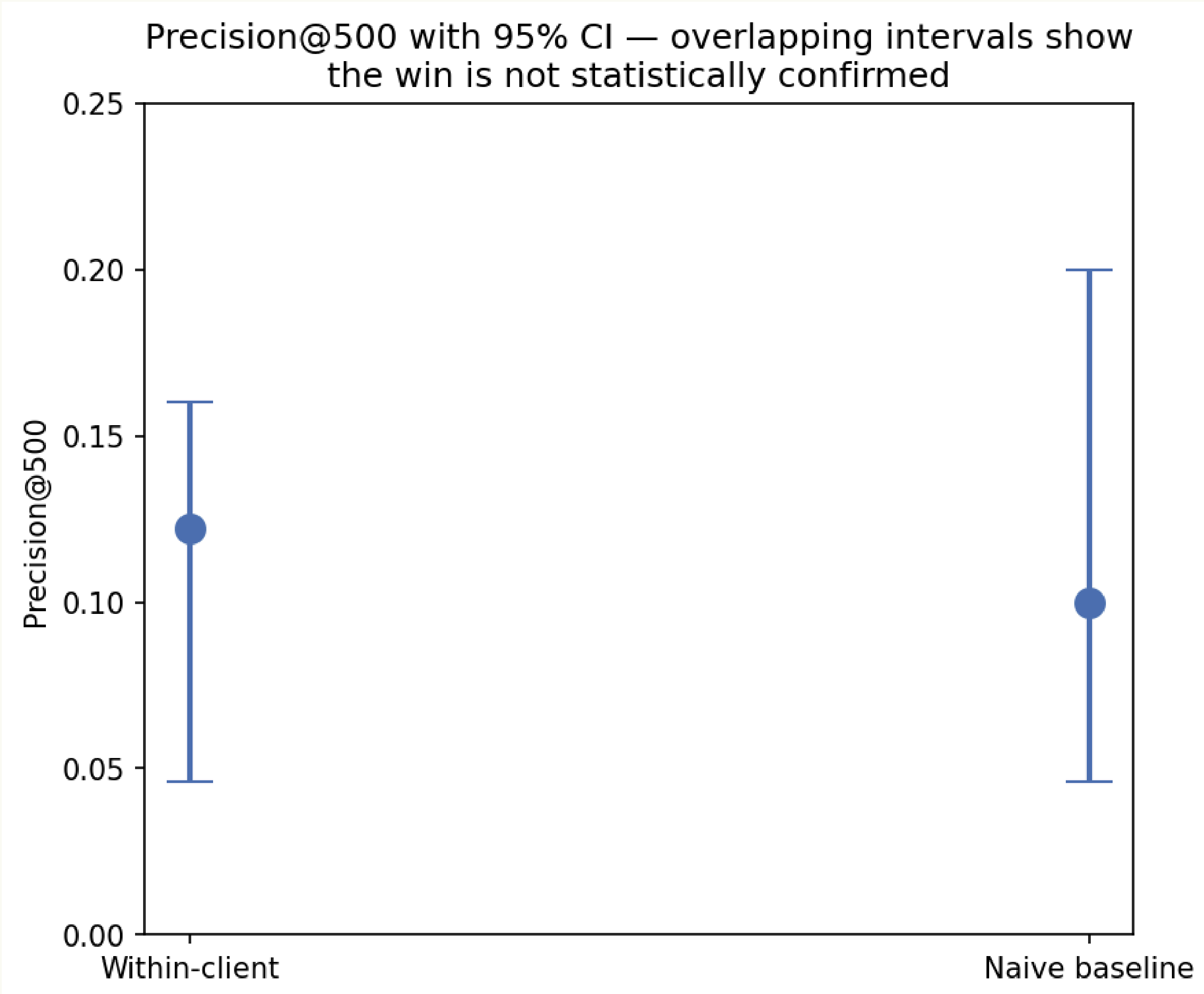


Figure 7 — *Chart read:* the two error bars overlap heavily — that overlap is the finding. A confirmed difference would show separated intervals; this doesn't, yet.

Removing the single most-represented client from the test set drops within-client precision below the naive baseline entirely — a reminder that this result rests on a thin client base (8 clients), not a fragile method.

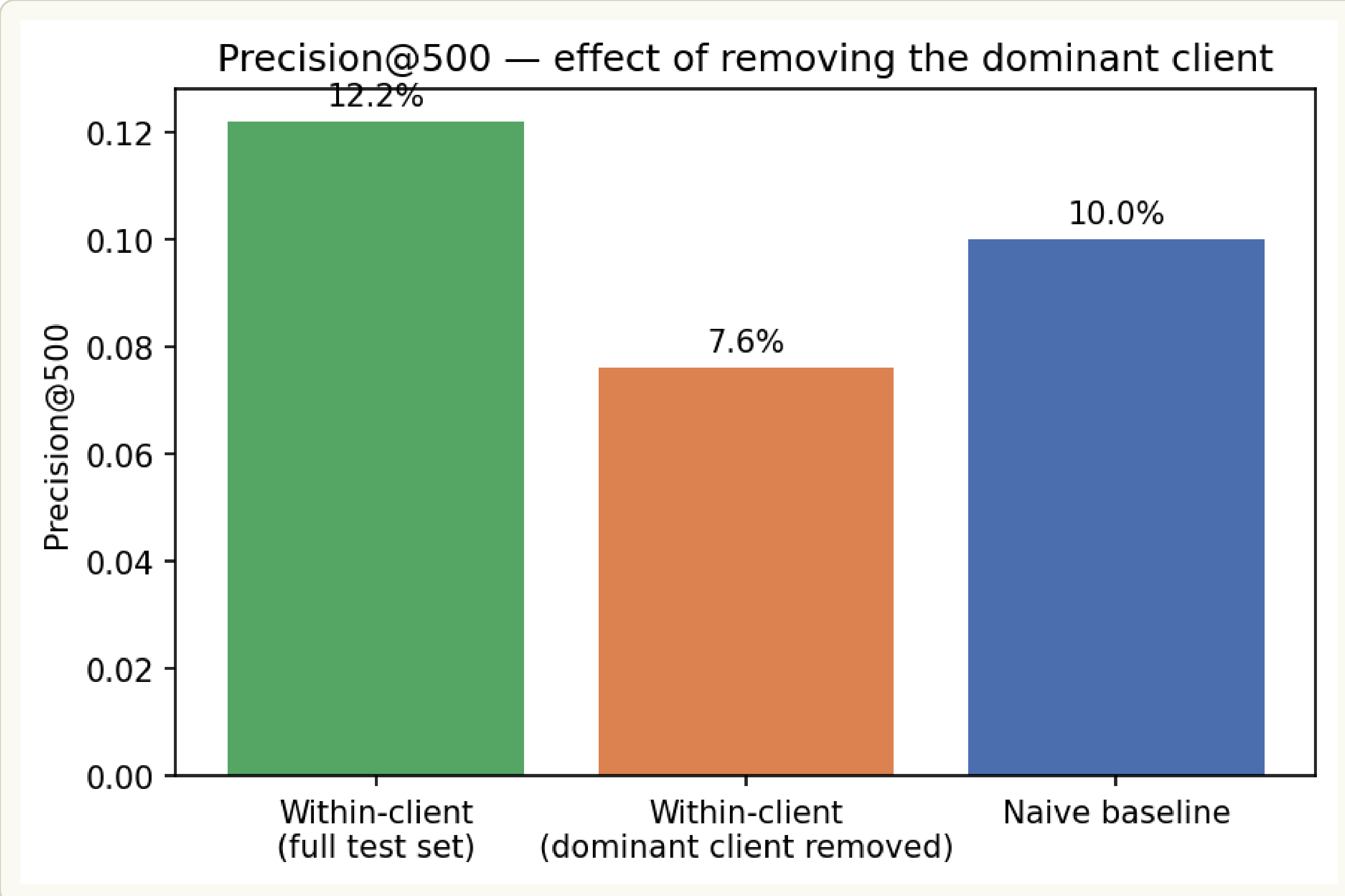


Figure 8 — *Chart read:* the orange bar (dominant client removed) falls below the naive baseline line — the win depends heavily on one client's volume.

Table 4. Bootstrap summary — precision, confidence intervals, and win margin.

METRIC	VALUE
Within-client precision@500 (point estimate)	12.2%
Within-client 95% CI	[4.6%, 16.0%]
Naive precision@500 (point estimate)	10.0%
Naive 95% CI	[4.6%, 20.0%]
Mean paired win margin (1,000 resamples)	**+0.34pp**
95% CI on win margin	[−4.60pp, +3.20pp]
Resamples where within-client wins	58.5%
Precision@500 without dominant client	7.6%

Analysis: the two headline precision figures overlap heavily in their individual 95% CIs (4.6–16.0% for within-client vs. 4.6–20.0% for naive), visually confirmed by Figure 7's overlapping intervals — but the paired bootstrap is the more informative test, since it holds the resample fixed and looks at the difference client-by-client rather than comparing two independent-looking intervals. That paired view shows the average edge is tiny: +0.34 percentage points, not the +2.2pp gap the single test-set snapshot suggests, and its own 95% CI still spans from a 4.60pp loss to a 3.20pp win. Only 58.5% of resamples favored the within-client score at all — barely better than a coin flip — and pulling the single most-represented client out of the test set drops within-client precision to 7.6%, below the naive baseline entirely. Together these numbers say the point-estimate win on the original test split likely owes more to which clients happened to land in the test group than to a robust property of the scoring method.

Recommendation: don't report the 8.76× lift or 12.2% precision figures as a confirmed win in any external-facing claim — they're accurate as point estimates on this specific split, but the paired-resample view shows they don't survive resampling with confidence. For internal use, treat the within-client score as a legitimate improvement in ranking logic (it fixes the client-concentration problem documented in Section 4) worth keeping, but gate any performance claim behind either a larger, more balanced test population (see Section 6) or a live validation where restructured high_opportunity pages are tracked against a held-out control group over a full quarter — a stronger test than a single retrospective split can provide.

Leakage check: injecting a feature derived from the label itself produced a 66.47× lift — 7.6× inflated versus the honest 8.76×. That gap confirms the evaluation pipeline correctly detects an obvious leak rather than silently accepting an inflated result, which is part of why the honest 8.76× figure above can be trusted.

SECTION 6

Limitations — what this cannot claim

Representativeness. Only 39 of 104 total clients (37.5%) are represented in the eligible feature set, and only 8 landed in the test split. Any finding here describes this 39-client subset, not FlyRank's client base as a whole.

Figure 9 — *Chart read:* green segments are clients that reach the eligible feature set; red segments are excluded — most exclusion happens before any modeling choice, by access profile alone.

Table 5. Client representativeness by access profile.

ACCESS PROFILE	CLIENTS (OF 104)	REACHING FACT TABLE
gsc_and_ga4	**53**	**42**
no_search_or_analytics_access	26	3
gsc_only	14	10
source_only_missing_client_dimension	10	4
ga4_only	1	0
Total	104	59

Analysis: access profile alone explains most of the client attrition before any modeling choice is made — the 26 clients with no search or analytics access at all contribute only 3 to the fact table, and the single ga4_only client contributes none, since GSC data specifically is what this capstone's features require. Even the best-case gsc_and_ga4 profile, the segment most complete, only gets 42 of 53 clients into the fact table; the remaining 59-to-39 gap (from the exclusion rules above — deleted articles, sub-14-day history) then trims that pool further to the 39 eligible clients used throughout Section 5–Section 7.

Recommendation: treat this table as a checklist for future data-quality work, not just a limitation note. The 23-client gap within no_search_or_analytics_access (26 total, only 3 reaching the fact table) is the single largest lever available to grow the eligible client base — larger than any of the other four segments combined. If FlyRank's data team can confirm and fix even half of that gap, the eligible pool would grow from 39 toward something closer to 50–55 clients, a meaningful step toward the much larger sample sizes discussed below that would be needed to move the headline result from directionally promising to statistically confirmed.

The primary result is not statistically distinguishable from the naive baseline at this test-set size, for the reasons shown in Section 4. With only 8 test clients, this ceiling can't be resolved by further bootstrapping or tuning — it needs a larger, more balanced client population to test honestly. Concretely: the bootstrap's own mean paired win margin is just +0.34pp, with a 95% CI of [−4.60pp, +3.20pp] — a lower-side gap of 4.94pp that would need to close relative to that same +0.34pp mean before the interval clears zero, a roughly 14.7× reduction in that gap. Since bootstrap CI width scales with $1/\sqrt{\text{(independent clients)}}$, closing it would take on the order of $14.7^2 \times 8 \approx 1{,}730$ test clients — a client base larger than FlyRank's entire current roster of 104 — before the result could move from "not confirmed" to "confirmed." That scale is itself informative: it suggests the true edge may be smaller and more client-dependent than the +2.2pp point estimate on this specific split implies, not simply "needs more data" in the usual sense.

The raw (non-within-client) composite actually underperforms the naive baseline — cross-client z-scoring lets high-volume clients dominate the ranking. That's a finding in its own right: raw composite scoring shouldn't be recommended without the within-client correction.

Train/test label-rate gap, only partially resolved. A single outlier client holds roughly half of all positive articles in the entire dataset and couldn't be split across train and test without breaking client-level leakage protection. Stratifying by label rate narrowed the train/test gap to 2.34% / 1.39% — a real gap remains, a structural ceiling of having only 39 clients, not a fixable split-logic issue.

Structural, unresolvable limits of this data release:

- No causal claims are possible — this measures association between content traits and AI-referral sessions, never why an AI system selected a page.
- Word count is missing for roughly 31% of articles regardless of window; any feature relying on it inherits that gap.
- The AI-referral signal measures post-click sessions attributed to AI referral, not which content elements an AI system actually used or valued.
- This is a single, non-seasonal 3-month window — no finding here can speak to seasonal effects or longer-term trend.
- 12 articles show real AI-referral activity despite zero measured search impressions — direct evidence the two search-derived features can't explain all AI-referral activity on their own.

Table 6. Articles with AI-referral activity despite zero GSC signal.

CONTENT	CLIENT	AI SESSIONS	WORD COUNT
content_abaa1fe…	client_4a18d17…	1	1,746
content_cad8c97…	client_4a18d17…	1	1,424
content_f4d79f4…	client_4a18d17…	3	5,140
content_b1a97a9…	client_625b643…	1	—
content_a6bbe85…	client_625b643…	1	—
content_9f3678c…	client_625b643…	1	—
content_d86846b…	client_625b643…	1	—
content_05e92b4…	client_4a18d17…	1	1,748
content_44a53f9…	client_4a18d17…	3	2,135
content_cf4609f…	client_4a18d17…	1	1,168
content_6fc9687…	client_4a18d17…	3	3,793
content_9c163f9…	client_625b643…	2	—
Total (12 articles)	**2 clients**	**19**	**7 of 12 present**

Analysis: all 12 articles trace back to just two clients, and both are members of the same zero-GSC-signal group flagged in Section 3's New Finding callout — client_4a18d17… (7 of the 12 articles) and client_625b643… (5 of 12). That's not a coincidence: these are articles with real, measured AI-referral sessions (1–3 each, 19 total) sitting inside accounts this capstone's eligibility filter currently labels `insufficient_data` and excludes from scoring entirely, purely because their GSC impressions are structurally zero. Five of the twelve are also missing word_count, so for those rows neither of the two features this score relies on is available — yet the AI-referral activity is real and measured via GA4, independent of GSC. This is direct, if small-sample, evidence that the two GSC-derived features can't be the whole story behind AI-referral pickup, since here it's happening with no GSC signal behind it at all.

Recommendation: don't score these 12 articles with the current within-client formula — there's no GSC-derived signal to score them on, and forcing a value would misrepresent confidence. Instead, flag them explicitly as a distinct "GSC-blind but AI-active" watchlist for manual review, and treat their existence as the strongest single argument in this paper for adding a GA4-native feature (e.g. a rolling AI-session count) in a future iteration, rather than continuing to score opportunity purely from search-console signals as done here.

None of this is a reason to discard the work. The leakage check in Section 4 shows the evaluation pipeline correctly distinguishes a real signal from an artificially inflated one, and the within-client score is directionally promising and mechanistically sound — it fixes a real concentration problem, and it happens to also lead on precision. What it lacks is statistical confirmation at this sample size, which is itself a legitimate, reportable finding, not a failure of the method.

SECTION 7

Ranked recommendations

This is a **candidate list for human review**, not a validated prediction. Every recommendation below should be read as "worth a content strategist's attention," not "will generate AI-referral traffic."

Who this is for: a content strategist deciding where to spend limited restructuring effort. **What "recommended" means:** an article that ranks highly on the within-client score — high search visibility relative to its own client's typical article, similar traits to articles that already receive AI-referral sessions — but currently has zero observed AI-referral activity, i.e. visible enough to be a plausible candidate, but not yet capturing that traffic.

Where every eligible article lands:

low_signal — 205,476 insufficient_data — 42,734 high_opportunity — 12,126

established_ai_source — 5,839

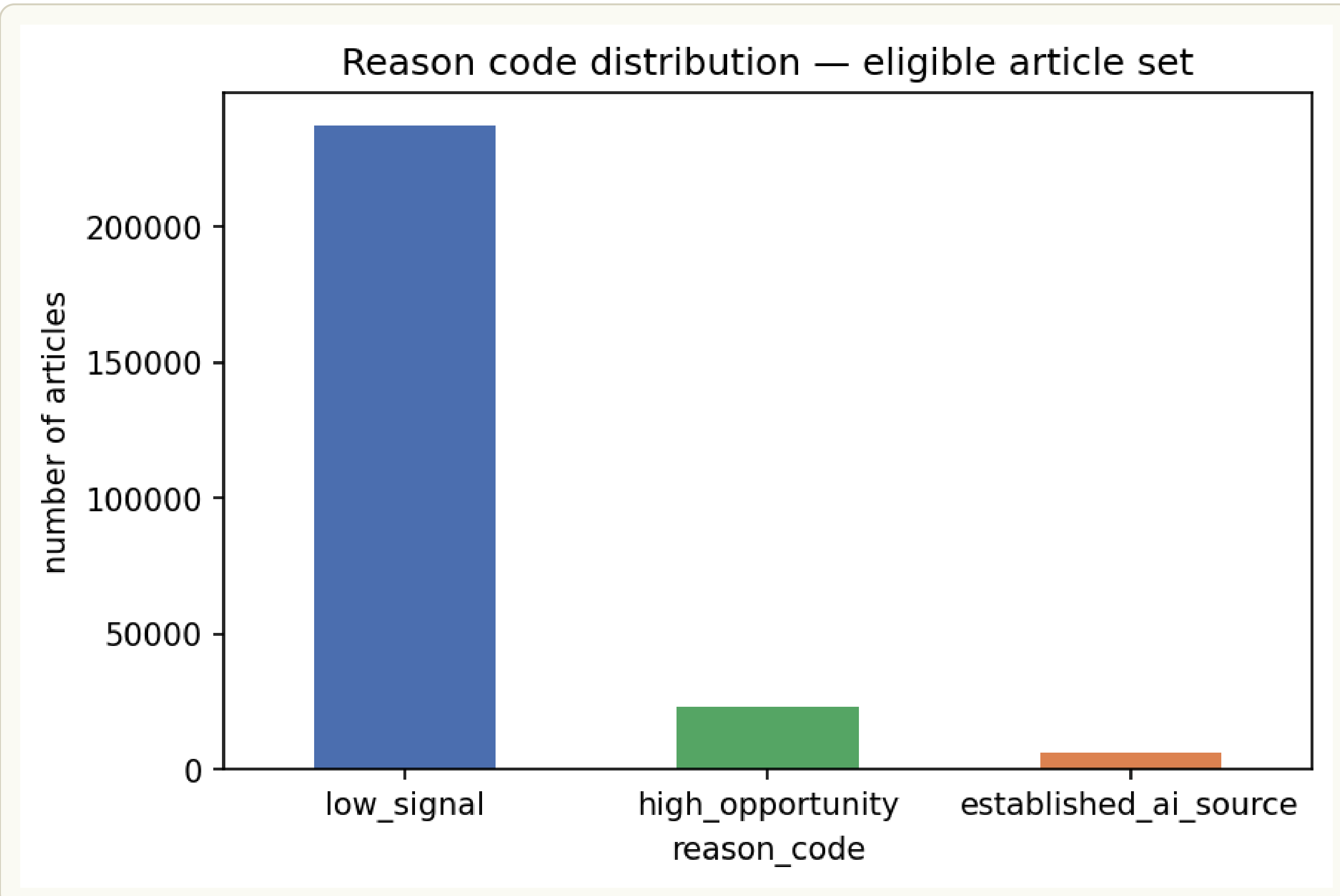


Figure 10 — *Chart read:* `low_signal` dominates by design — most articles simply aren't standout candidates in any given window.

Of the 12,126 `high_opportunity`-tagged articles, the final exported list caps at 10 per client to prevent any one client's volume from dominating the output — **335 candidates across 35 clients.** An earlier version of this list let zero-impression articles through on word count alone, contradicting the lane's own premise that a candidate must already have search visibility; it was fixed by requiring impressions above the client's own median before qualifying.

Top of the exported queue — the eight highest within-client scores, straight from `capstone_ranked_recommendations.csv`. Hash keys are truncated and salted per the release's own anonymization; word count shows – where it's part of the 31% structurally missing.

CAPSTONE_RANKED_RECOMMENDATIONS.CSV — SORTED BY SCORE, DESC		336 ROWS TOTAL
1	client_3ffa763… / content_1e3fe8… 2,873 impr · word count –	score 51.08
2	client_3ffa763… / content_67ae61… 2,479 impr · word count –	score 44.07
3	client_73cda7b… / content_fec559… 118,018 impr · word count –	score 39.15
4	client_f623b01… / content_b5bef9… 22,438 impr · 1,108 words	score 25.53
5	client_3ffa763… / content_025a92… 1,393 impr · word count –	score 24.72
6	client_62f4a7e… / content_acbcc8… 202,211 impr · word count –	score 23.98

7	client_cd12bcf… / content_99fc64… 5,084 impr · 3,338 words	
score 23.74		
8	client_73cda7b… / content_8e1334… 71,863 impr · word count –	
score 23.70		

How to use this list:

- Review `high_opportunity` candidates for restructuring — clearer definitions, more explicit answer framing — treating each as worth a look, not a guaranteed lift, given Section 5's significance finding.
- Treat `established_ai_source` pages as reference examples of what "already working" looks like for that client, not action items.

What this list explicitly does not claim: that restructuring a `high_opportunity` article will increase its AI-referral sessions. It claims only that the article resembles, on the two features measured, articles that already do receive them — an association, not a mechanism.

SECTION 8

Reproducibility

Everything above is built and verified in the capstone notebook, in the same 9-section order as this paper, in the project repository:

- **Capstone notebook:** work/notebooks/capstone.ipynb
- **Full repository:** github.com/whozahm3d/flyrank-ml-internship
- **Consolidated findings reference:** work/SUMMARY.md
- **PDF version of this paper:** ai-referral-opportunity-scoring.pdf

Randomness & environment: the client-grouped train/test split and every bootstrap resample in Section 5 use a fixed random seed of `42` (`numpy.random.default_rng(42)`), so the split and the reported confidence intervals reproduce exactly on re-run. The notebook was run against `pandas>=2.2`, `numpy>=1.26`, and `scikit-learn>=1.4`, per the repository's requirements.txt.

Bootstrap methodology: confidence intervals in Section 5 are computed with a percentile bootstrap resampled at the *client* level, not the article level — for each of 1,000 iterations, the 8 test clients are resampled with replacement and precision@500 is recomputed across all articles belonging to the resampled clients, then the 2.5th and 97.5th percentiles of the resulting distribution form the reported 95% CI. Resampling at the client level (rather than pooling all articles and resampling individually) is what makes 8 the effective sample size driving CI width, not 40,631 — articles from the same client aren't independent observations for this purpose.

Glossary

Precision@K	Of the top K articles by score, the share that actually went on to receive real AI-referral sessions.
Lift@K	Precision@K divided by the dataset's overall positive rate (2.20%) — how much better than random selection the score does at that K.
pp	Percentage points — the arithmetic difference between two percentages (e.g. 12.2% − 10.0% = 2.2pp), not a relative percent change.
Bootstrap CI	A confidence interval built by resampling the data thousands of times and taking the spread of results, rather than assuming a textbook distribution — see the methodology note above for exactly how it's computed here.
Within-client normalization	Scoring each article relative to its own client's typical article, instead of comparing raw values across clients of very different size — the fix described in Section 4.
Leakage / leakage check	When information from the future or from the label itself accidentally informs a score, inflating results; the leakage check in Section 5 deliberately injects a label-derived feature to confirm the pipeline catches it.

Every raw output cited in this paper, linked to the number or chart it backs — all in work/outputs/:

Table 7. Reproducibility file index — each output CSV and what it backs.

FILE	BACKS
capstone_ranked_recommendations.csv	Section 7 — the 336-row ranked queue and its top-8 excerpt
reason_code_distribution.csv	Section 7 — the funnel bar and chart (205,476 / 42,734 / 12,126 / 5,839)
results_lift_precision.csv	Section 5 — the precision@500 / lift@500 comparison table
results_lift_robustness.csv	Section 5 — lift across K=100 to K=2,000
results_summary_stats.csv	Section 5 — the bootstrap CI [−4.60pp, +3.20pp] and the 58.5% win-rate figure
bootstrap_paired_resamples.csv	Section 5 — the underlying 1,000 paired client-level resamples behind that CI
client_access_representativeness.csv	Section 6 — the 39/104-client representativeness chart
client_stats.csv	Section 6 — per-client article counts and label rates behind the representativeness and dominant-client findings
zero_gsc_clients.csv	Section 3 — the 6-client, 42,734-row eligibility-filter gap (New finding callout)
ai_sessions_without_gsc_signal.csv	Section 6 — the 12 articles with real AI-referral activity despite zero measured impressions
summary_verdicts.csv	Week-4 signal audit verdicts that shaped which features made it into the score (background context, not directly cited above)

Three more files in that folder — `w07_baseline_history.jsonl`, `w07_baseline_snapshot.json`, `w07_export_manifest.json` — belong to the separate content-decline lane described below, not this capstone.

This capstone builds directly on five earlier notebooks in the same repository — lane definition and starter-sample findings, the original opportunity-score composite, the warehouse data contract (including the first client-metadata staleness finding), and a dedicated leakage-detection pass.

A separate, unrelated curriculum track in the same repository (five additional notebooks, `work/notebooks/w04–w07`) developed a content-decline lane on the same 30,000-row starter dataset — a different research question, method, and label from AI-referral opportunity, with no component carried into this capstone's features, scoring, or findings.

> **How to cite this work:** Ahmad, Ali. (2026). *Which pages are worth restructuring for AI-referral traffic? — AI Referral Opportunity Scoring.* FlyRank ML Engineering Internship Capstone Research Paper. github.com/whozahm3d/flyrank-ml-internship.

SECTION 9

Acknowledgments & data credit

Built on the **FlyRank ML Internship dataset**.

Thanks to the FlyRank ML track curriculum for structuring this work week by week, and to the internship warehouse release for making a real, large-scale search-intelligence dataset available for hands-on learning.

SECTION 10

Contacts

Questions, corrections, or collaboration inquiries about this paper are welcome through any of the channels below.

Based in Sialkot, Pakistan (UTC+5). Email is the best channel for detailed questions or collaboration proposals — LinkedIn and X work well for quicker professional notes.